

Algebraic Techniques

Week 2A

Overview

Algebraic Fractions

Key Ideas - Simplifying Algebraic Fractions

- Multiplying Algebraic Fractions

- Dividing Algebraic Fractions

- Adding and Subtracting Algebraic Fractions

Key Idea – Simplifying Algebraic Fractions

An algebraic fraction is **a fraction with algebraic expressions in the denominator and/or numerator**, for example:

$$\frac{2x^2 + y}{6xy}$$
$$\frac{1}{a^2 - b^2}$$
$$\frac{ax^2 - q^3}{y}$$

Manipulation of algebraic fractions is one of the most error-prone Algebraic techniques – but it involves vital skills that **MUST** be mastered.

Remember, variables are just place-holder names for unknown numbers. So, the **"rules" for algebraic fractions are the same as the "rules" for ordinary fractions.**

We can simplify numerical fractions by cancelling factors that are common to the numerator and the denominator. This rule also applies to algebraic fractions.

Discussion Questions:

Simplify the fraction $\frac{15}{20} = \frac{\cancel{3} \times \cancel{5}}{\cancel{4} \times \cancel{5}}$
 $= \frac{3}{4}$

Could you use this same approach to simplify a fraction such as $\frac{3x^2}{21x}$?

Yes $\frac{3x^2}{21x} = \frac{\cancel{3} \times \cancel{x} \times x}{7 \times \cancel{3} \times \cancel{x}}$
 $= \frac{x}{7}$

Important skills are required for the successful manipulation of algebraic fractions, these skills include, knowing how to **factorise algebraic expressions** and a knowledge of calculating numerical fractions.

There are three steps involved in simplifying algebraic fractions:

- **Factorise the numerator and the denominator.**
- **Identify common factors** between the numerator and the denominator.
- **Cancel out common factors** in the numerator and denominator.

***Note:** A common factor is a factor that occurs in both the numerator and the denominator.*

Examples: Simplify the following:

$$\frac{12ab}{18db} = \frac{\cancel{2} \times 2 \times \cancel{3} \times a \times \cancel{b}}{\cancel{2} \times \cancel{3} \times 3 \times d \times \cancel{b}} = \frac{2a}{3d}$$

$$\frac{10b^3c}{5b^2c^2} = \frac{2 \times \cancel{5} \times \cancel{b} \times \cancel{b} \times b \times \cancel{c}}{\cancel{5} \times \cancel{b} \times \cancel{b} \times \cancel{c} \times c} = \frac{2b}{c}$$

$$\frac{5a-15}{a^2-9} = \frac{5(\cancel{a-3})}{(\cancel{a-3})(a+3)} = \frac{5}{a+3}$$

$$\frac{(a+b)^2 - c^2}{2a+2b-2c} = \frac{(\cancel{a+b-c})(a+b+c)}{2(\cancel{a+b-c})} = \frac{(a+b+c)}{2}$$

$$\frac{x^2-10x+21}{x^2-2x-3} = \frac{(x-7)(\cancel{x-3})}{(\cancel{x-3})(x+1)} = \frac{(x-7)}{(x+1)}$$

Exercices: Simplify the following algebraic fractions:

$$(a) \quad \frac{8m-16}{m^2-4m+4} = \frac{\cancel{8(m-2)}}{\cancel{(m-2)}(m-2)} \\ = \frac{8}{m-2}$$

$$(b) \quad \frac{a^3-27}{a^2-9} = \frac{a^3-3^3}{a^2-3^2} \\ = \frac{\cancel{(a-3)}(a^2+3a+9)}{\cancel{(a-3)}(a+3)} \\ = \frac{a^2+3a+9}{a+3}$$

Key Idea – Multiplication of Algebraic Fractions

Multiplication of algebraic fractions follows the same process as multiplication of numerical fractions.

Discussion Questions:

How do we multiply fractions such as $\frac{2}{3} \times \frac{7}{10}$ without a calculator?

$$\begin{aligned}\frac{2}{3} \times \frac{7}{10} &= \frac{2 \times 7}{3 \times 10} \\ &= \frac{\cancel{2} \times 7}{3 \times 5 \times \cancel{2}} = \frac{7}{15}\end{aligned}$$

Could you use this same approach to multiply the fractions $\frac{6m}{5a} \times \frac{15a}{2m}$?

$$\begin{aligned}\text{Yes} \quad \frac{6m}{5a} \times \frac{15a}{2m} &= \frac{6m \times 15a}{5a \times 2m} \\ &= \frac{\cancel{3} \times \cancel{2} \times \cancel{m} \times \cancel{3} \times \cancel{5} \times \cancel{a}}{\cancel{5} \times \cancel{a} \times \cancel{2} \times \cancel{m}} \\ &= 9\end{aligned}$$

There are four steps involved in multiplying algebraic fractions:

- **Factorise the numerator and denominator** of each fraction.
- Write factors of both numerators together as the new numerator and write factors of both denominators together as the new denominator. (This step can be omitted).
- **Cancel common factors** between the new numerator and new denominator.
- **Multiply across** the numerator and multiply across the denominator

Examples: Simplify the following:

$$\begin{aligned}\frac{3ab}{4a} \times \frac{8a^2}{15b} \times \frac{1}{b} &= \frac{3ab \times 8a^2 \times 1}{4a \times 15b \times b} \\ &= \frac{\cancel{3} \times \cancel{a} \times \cancel{b} \times 2 \times \cancel{4} \times a \times a \times 1}{\cancel{4} \times \cancel{a} \times 5 \times \cancel{3} \times \cancel{b} \times b} \\ &= \frac{2a^2}{5b}\end{aligned}$$

$$\begin{aligned}\frac{x-2}{x+2} \times \frac{3x+6}{x^2+2x-8} &= \frac{x-2}{x+2} \times \frac{3(x+2)}{(x+4)(x-2)} \\ &= \frac{3 \times \cancel{(x-2)} \times \cancel{(x+2)}}{\cancel{(x+2)} \times (x+4) \times \cancel{(x-2)}} \\ &= \frac{3}{(x+4)}\end{aligned}$$

Exercises: Simplify the following fractions:

$$\begin{aligned} \text{(a)} \quad \frac{x^2 - 7x + 10}{x^2 - 10x + 25} \times \frac{x - 5}{x - 2} &= \frac{(x-5)(x-2)}{(x-5)(x-5)} \times \frac{(x-5)}{(x-2)} \\ &= \frac{\cancel{(x-5)}\cancel{(x-2)}\cancel{(x-5)}}{\cancel{(x-5)}\cancel{(x-5)}\cancel{(x-2)}} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \frac{a-b}{8} \times \frac{16}{b^2 - a^2} &= \frac{(a-b) \times 16}{8 \times (b^2 - a^2)} \\ &= \frac{(a-b) \times 2 \times \cancel{8}}{\cancel{8} \times (b-a)(b+a)} \\ &= \frac{\cancel{(a-b)} \times 2}{-\cancel{(a-b)}(b+a)} \\ &= -\frac{2}{b+a} \end{aligned}$$

Key Idea – Division of Algebraic Fractions

Division by an algebraic fraction follows the same process as division by a numerical fraction.

Discussion Questions:

How do you divide fractions such as $\frac{1}{2} \div \frac{7}{10}$ without a calculator?

$$\frac{1}{2} \div \frac{7}{10} = \frac{1}{2} \times \frac{10}{7}$$

$$= \frac{1 \times 10}{2 \times 7}$$

$$= \frac{1 \times \cancel{2} \times 5}{\cancel{2} \times 7} = \frac{5}{7}$$

Could you use this same approach to divide the fractions $\frac{10k}{3n} \div \frac{2k}{9n}$?

Yes

$$\frac{10k}{3n} \div \frac{2k}{9n} = \frac{10k}{3n} \times \frac{9n}{2k}$$

$$= \frac{10k \times 9n}{3n \times 2k}$$

$$= \frac{5 \times \cancel{2} \times \cancel{k} \times \cancel{3} \times 3 \times n}{\cancel{3} \times n \times \cancel{2} \times \cancel{k}} = 15$$

There are four steps involved in dividing by an algebraic fraction:

- **Take the reciprocal of the second fraction** and multiply.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

- **Factorise** the numerator and denominator of each fraction.
- **Cancel** common factors between the new numerator and the new denominator.

Multiply across the numerator and multiply across the denominator

Examples: Simplify the following:

$$\begin{aligned} \frac{3b}{c} \div \frac{9b^3a}{d} &= \frac{3b}{c} \times \frac{d}{9b^3a} \\ &= \frac{3b \times d}{c \times 9b^3a} \\ &= \frac{\cancel{3} \times \cancel{b} \times d}{c \times \cancel{3} \times 3 \times \cancel{b} \times b \times b \times a} \\ &= \frac{d}{3ab^2c} \end{aligned}$$

$$\begin{aligned} \frac{y^2 - y}{3y^3 + 3y^2} \div \frac{y^2 - 1}{6y} &= \frac{y^2 - y}{3y^3 + 3y^2} \times \frac{6y}{y^2 - 1} \\ &= \frac{y(y-1)}{3y^2(y+1)} \times \frac{6y}{(y-1)(y+1)} \\ &= \frac{\cancel{y} \times \cancel{(y-1)} \times 2 \times \cancel{3} \times \cancel{y}}{\cancel{3} \times \cancel{y} \times \cancel{y} \times (y+1) \times \cancel{(y-1)} \times (y+1)} \\ &= \frac{2}{(y+1)^2} \end{aligned}$$

Exercises: Simplify the following fractions:

$$\begin{aligned} \text{(a)} \quad \frac{15}{3x-6} \div \frac{3}{9x-18} &= \frac{15}{3x-6} \times \frac{9x-18}{3} \\ &= \frac{3 \times 5}{3(x-2)} \times \frac{9(x-2)}{3} \\ &= \frac{\cancel{3} \times 5 \times \cancel{3} \times 3 \times \cancel{(x-2)}}{\cancel{3} \times \cancel{(x-2)} \times \cancel{3}} \\ &= 15 \end{aligned}$$

Exercises: Simplify the following fractions:

$$\begin{aligned} \text{(b)} \quad \frac{x^2 - x - 20}{x^2 - 25} \div \frac{x+1}{x^2 + 5x} &= \frac{x^2 - x - 20}{x^2 - 25} \times \frac{x^2 + 5x}{x+1} \\ &= \frac{(\cancel{x-5})(x-4)}{(\cancel{x-5})(\cancel{x+5})} \times \frac{x(\cancel{x+5})}{(x+1)} \\ &= \frac{x}{x+1} \end{aligned}$$

Exercises: Simplify the following fractions:

$$\begin{aligned} \text{(c)} \quad \frac{\frac{a-2b}{ab^3}}{\frac{2b-a}{a^3b}} &= \frac{a-2b}{ab^3} \div \frac{2b-a}{a^3b} \\ &= \frac{a-2b}{ab^3} \times \frac{a^3b}{2b-a} \\ &= \frac{(a-2b) \times \cancel{a} \times a \times a \times \cancel{b}}{\cancel{a} \times \cancel{b} \times b \times b \times (2b-a)} \\ &= \frac{-(\cancel{2b-a}) \times a \times a}{b \times b \times \cancel{(2b-a)}} \\ &= -\frac{a^2}{b^2} \end{aligned}$$

Key Idea – Addition and Subtraction of Algebraic Fractions

Addition and subtraction of algebraic fractions follow the same process as addition and subtraction of numerical fractions.

Discussion Questions:

How do we perform addition of numerical fractions such as $\frac{1}{5} + \frac{2}{3}$?

$$\begin{aligned}\frac{1}{5} + \frac{2}{3} &= \frac{1 \times 3}{5 \times 3} + \frac{2 \times 5}{3 \times 5} \\ &= \frac{3}{15} + \frac{10}{15} \\ &= \frac{13}{15}\end{aligned}$$

Could you use this same approach to add the fractions $\frac{1}{5x} + \frac{2}{3x}$?

$$\begin{aligned}\frac{1}{5x} + \frac{2}{3x} &= \frac{1 \times 3}{5x \times 3} + \frac{2 \times 5}{3x \times 5} \\ &= \frac{3}{15x} + \frac{10}{15x} \\ &= \frac{13}{15x}\end{aligned}$$

The rules for addition and subtraction of algebraic fractions are:

- If possible, factorise the denominator of each fraction to find the **smallest common denominator**.
- **Convert each fraction**, so that they all have a common denominator.
- **Add or subtract the numerators**.

Note: $\frac{x}{z} + \frac{y}{z} = \frac{x+y}{z}$

Examples: Simplify the following:

$$\begin{aligned}\frac{x+3}{2} + \frac{x-1}{3} &= \frac{(x+3) \times 3}{2 \times 3} + \frac{(x-1) \times 2}{3 \times 2} \\ &= \frac{3(x+3) + 2(x-1)}{6} \\ &= \frac{3x+9+2x-2}{6} \\ &= \frac{5x+7}{6}\end{aligned}$$

$$\begin{aligned}\frac{2x}{3a} - \frac{5x}{3b} &= \frac{2x}{3a} \times \frac{b}{b} - \frac{5x}{3b} \times \frac{a}{a} \\ &= \frac{2bx}{3ab} - \frac{5ax}{3ab} \\ &= \frac{2bx-5ax}{3ab} \\ &= \frac{(2b-5a)x}{3ab} \quad (\text{this step is optional})\end{aligned}$$

Examples: Simplify the following:

$$\begin{aligned}\frac{5}{a} - \frac{4}{a^2 - a} &= \frac{5}{a} - \frac{4}{a(a-1)} \\&= \frac{5}{a} \times \frac{(a-1)}{(a-1)} - \frac{4}{a(a-1)} \\&= \frac{5(a-1) - 4}{a(a-1)} \\&= \frac{5a - 9}{a(a-1)}\end{aligned}$$

$$\begin{aligned}\frac{2}{x^2 - 4} - \frac{3}{(x-2)^2} &= \frac{2}{(x-2)(x+2)} - \frac{3}{(x-2)(x-2)} \\&= \frac{2}{(x-2)(x+2)} \times \frac{(x-2)}{(x-2)} - \frac{3}{(x-2)(x-2)} \times \frac{(x+2)}{(x+2)} \\&= \frac{2(x-2) - 3(x+2)}{(x+2)(x-2)^2} \\&= \frac{2x - 4 - 3x - 6}{(x+2)(x-2)^2} \\&= \frac{-x - 10}{(x+2)(x-2)^2}\end{aligned}$$

Exercises: Simplify the following fractions :

$$\begin{aligned} \text{(a)} \quad \frac{9}{x-2} + \frac{4}{x^2-3x+2} &= \frac{9}{(x-2)} + \frac{4}{(x-2)(x+1)} \\ &= \frac{9(x+1)}{(x-2)(x+1)} + \frac{4}{(x-2)(x+1)} \\ &= \frac{9(x+1)+4}{(x-2)(x+1)} \\ &= \frac{9x+9+4}{(x-2)(x+1)} \\ &= \frac{9x+13}{(x-2)(x+1)} \end{aligned}$$

Exercises: Simplify the following fractions :

$$\begin{aligned} \text{(b)} \quad \frac{3}{x^2 - 4} - \frac{5}{3x - 6} &= \frac{3}{(x-2)(x+2)} - \frac{5}{3(x-2)} \\ &= \frac{3}{(x-2)(x+2)} - \frac{5(x+2)}{3(x-2)(x+2)} \\ &= \frac{3 - 5(x+2)}{(x-2)(x+2)} \\ &= \frac{3 - 5x - 10}{(x-2)(x+2)} \\ &= \frac{-5x - 7}{(x-2)(x+2)} \\ &= \frac{-(5x + 7)}{(x-2)(x+2)} \end{aligned}$$

Why we can't Simplify some Fractions

Mistake 1

You **cannot** add or subtract the same thing in both top and bottom of a fraction—you can only **multiply top and bottom by 1**.

Example:

For $\frac{axy + 7}{bc + 7}$ it is tempting to “cancel” the 7's, but $\frac{axy + 7}{bc + 7} \neq \frac{axy}{bc}$.

If in doubt, try some examples with numbers: $\frac{20 + 7}{2 + 7} = \frac{27}{9} = 3$, but $\frac{20}{2} = 10$

In general, for $\frac{a + m}{b + m}$ the 'm' cannot be cancelled. That is $\frac{a + m}{b + m} \neq \frac{a}{b}$ nor $\frac{a + m}{b + m} \neq \frac{a + 1}{b + 1}$

Order of operations rules prohibit this step!!!!

You must calculate the addition first because: $\frac{a + m}{b + m} = (a + m) \div (b + m)$

Mistake 2

You **cannot** “cancel” something from just one term in the numerator or denominator.

Example:

For $\frac{y+a}{yb}$ it is tempting to “cancel” the y ’s, but $\frac{y+a}{yb} \neq \frac{1+a}{b}$.

This is because we did **not** multiply the **whole** fraction by y , just one of the two terms in the numerator.

If we had, the fraction $\frac{y+ay}{yb}$ we would be able to cancel out the y ’s using the following method:

$$\frac{y+ay}{yb} = \frac{\cancel{y} \times (1+a)}{\cancel{y} \times b}$$